

A Comparative Audit of Fundamental Theory Programmes

The Rope Programme alongside the Standard Model, general relativity, and the substrate-theory field. July 2026. Supersedes "A Scoreboard for Theories" on external review; the change from scoreboard to audit framework reflects the content, which is methodological.

Disclosure and method, stated before any scoring

This document is written by the team that built one of the programmes it evaluates: an audit by the home team. That conflict of interest cannot be removed, only disciplined. The discipline: comparison axes are fixed here, before scoring; entries are COUNTABLE FACTS wherever possible, so that readers verify rather than trust; the mainstream stack is scored first and placed where the evidence puts it — at the top; the Rope Programme's registered losses appear at the same prominence as its wins; and an appendix specifies exactly how to disagree with any entry. Where a metric cannot be computed for a theory because no public claim registry exists, the cell says so — and that absence is data about auditability, not about merit.

1. The champion: the Standard Model, quantum field theory, and general relativity

Any honest audit begins here, because the strongest competitor to every speculative programme is the mainstream stack itself. Electromagnetism and the weak interaction literally unified; the strong interaction described; all known matter organized; the electron's anomalous magnetic moment agreeing with calculation at roughly one part in a trillion; general relativity verified to a few parts in 10^5 in the solar system and by gravitational-wave observation in the strong field. Its audited breadth approaches its claimed breadth — the distinguishing glory no rival shares. Known incompleteness, equally on the record: nineteen-plus free parameters (twenty-six with neutrino masses and mixings); QFT–GR incompatibility; singularities; the cosmological constant discrepancy; the measurement problem; and no mechanism by design — the stack computes without saying what a field, a particle, or spacetime is.

2. The comparison field, briefly

STRING THEORY: maximal claimed breadth (all forces, all matter, from one-dimensional objects); mathematics of extraordinary depth; after five decades, no confirmed novel prediction and no principle selecting among $\sim 10^{500}$ discrete vacua. LOOP QUANTUM GRAVITY: a careful quantization of geometry, deliberately narrow; no confirmed observational contact. BOHMIAN MECHANICS: reproduces all of nonrelativistic quantum mechanics exactly — it strictly beats the Rope Programme on rope's weakest sector, stated plainly — while not attempting substrate unification of the forces. WOLFRAM HYPERGRAPHS: the Rope Programme's closest rival in ambition; on present public evidence its claims are structural rather than numerical, with no registered quantitative loss known to us; the axis-by-axis comparison this document most invites. VOLOVIK / EMERGENT PHYSICS: the best cousin — emergent gauge fields, chiral fermions, and effective metrics demonstrated in laboratory superfluid helium-3, offered honestly as analogy rather than fundamental theory; the tradition in which 'emergent metrics come easy, Einstein dynamics does not' was first learned.

3. The Rope Programme: countable facts from its public registry

The following numbers are computed from the machine-verified claim registry shipped with the corpus (claims.yaml, verify_corpus.py) and can be recomputed by any reader. Registered claims (as of the current release): 518. Benchmark-backed (executable, deterministic, currently passing): 497. By status: Derived 111; Modeled 369; EFT-constrained 4; Conjecture 4; Open 5; FAILED AND RETAINED 25. Absolute constants taken as inputs: approximately eight (mesh scale, $\hbar$, electron mass, nucleon unit, the stretch modulus's value, the vacuum tension density, G , the nuclear bond depth), several proven irreducible from within. Quantitative confrontations with published measurement, enumerated so the count is checkable: (1) PVLAS vacuum birefringence — LOSS, excluding the framework's identification by $\sim 570\times$; (2) ATLAS light-by-light — bound; (3) solar light deflection and Cassini — the direct-mechanism route was falsified and kept (0.44 vs 1.75 arcseconds), then superseded: the induced-gravity route DERIVES the weak-field metric, $\gamma = 1$, and the exact 1.751-arcsecond deflection at zero free parameters, and reproduces the SPARC galaxy acceleration relation; (4) the H_2 vibrational quantum — parameter-free, 16%; (5) atomic masses across the table from the light nuclei ($A \sim 8$) through U-238 — now a full semi-empirical mass formula derived from classical physics (volume, surface, Coulomb with diffuseness and exchange, asymmetry, and pairing terms), one calibrated constant, heavy-table binding to $\sim 1\%$ (the lightest nuclei remain the model's weakest region, and He-4 is the calibration anchor); (6) the liquid-drop surface coefficient — parameter-free geometry; the originally-cited 2% was found to be a facet formula misapplied to a sphere (NUC-016) and is honestly $\sim 18\%$ for the object modelled, later reformulated on a finite-range bond kernel giving $a_S/a_V \sim 1.26$ (NUC-D); (7) the Coulomb coefficient — the declared $+16\%$ miss closed to in-window by two derived classical corrections at zero new constants (diffuseness and a winding-correlation exchange hole, NUC-C); (8) the deuteron bond-depth direction check; (9) the nuclear composition rule against measured masses. Precision is SECTOR-DEPENDENT AND NOWHERE NEAR QFT: best 0.014%, typical 0.1–2%, against the mainstream's parts per trillion.

3a. Inherited versus independent precision (added on reader question)

A natural question about any programme that derives the Maxwell structure: does it not then predict electricity and magnetism to the same accuracy as Maxwell's equations — the same number of decimal places? The answer is yes, and the yes is cheaper than it sounds. Deriving the SAME equations makes every downstream prediction identical not to some number of matching decimals but EXACTLY, because it is the same mathematics: in the classical regime, the Rope Programme's electromagnetic predictions ARE Maxwell's, by inheritance. (In fairness, classical Maxwell theory likewise takes e and ϵ_0 as measured inputs, so the constants ledgers of the two classical accounts are comparable.) But inherited precision is not independent verification: Maxwell's parts-per-billion record was earned by 160 years of experiments testing those equations, and reproducing the equations inherits the record while generating no new numbers that could lose. This is precisely why Section 3 counts nine experimental confrontations rather than thousands — agreement-by-derivation is not a confrontation, and counting it as one would be the exact claim-inflation this framework forbids. Moreover, the celebrated modern decimals — the electron's magnetic moment at one part in a trillion — are not Maxwell at all but QED, requiring $\hbar$ and machinery the programme has declared out of scope. And at the first point where the programme's electromagnetism goes BEYOND linear Maxwell — its quartic vacuum nonlinearity — it disagrees with QED's (3:1 anisotropy with negative index shifts versus 7:4

positive), and in the one confrontation to date (PVLAS) experiment excluded the programme's initial identification by a factor of ~570. Summary: identical to Maxwell wherever Maxwell is exact, by inheritance; beyond Maxwell, either out of scope or making its own falsifiable — and once falsified — predictions. This is why the audit table's precision cell reads as it does.

4. The audit table

Programme	Public claim registry	Executable benchmarks	Quantitative confrontations	Registered failures kept	Adjustable parameters	Physical substrate specified
SM + QFT + GR	de facto (literature)	yes (community codes)	thousands; to 1e-12	exemplary record	19+ (26 w/ neutrinos) + G, Lambda	no (by design)
String theory	none	none public	0 confirmed	none callable	~10 ⁵⁰⁰ discrete vacua	partial (strings, not spacetime)
Loop quantum gravity	none	limited research codes	0 confirmed	none yet	few (Immirzi)	geometry only
Bohmian mechanics	n/a (= QM empirically)	= QM's	= QM's, exactly	shares QM's record	= QM's	yes (nonlocal particles + wave)
Wolfram hypergraphs	none found	framework code, not claim-tests	none found by us	none found	rule choice unspecified	yes (hypergraph)
Volovik / emergent	n/a (analogy scope)	laboratory	many, in-lab	standard lab record	material params (measured)	yes (superfluid)
Rope Programme	yes: 121 claims, statuses public	107/107 passing, shipped	9 enumerated (Sec. 3); best 0.014%; NOWHERE NEAR QFT	7 Failed kept + 2 losses + 1 refused rescue	~8 absolute inputs; forms/ratios parameter-free	yes (tensioned filament network)

Reading rules: cells are countable facts or explicit 'none found' statements; 'none' in the registry and benchmark columns means the metric is UNCOMPUTABLE for that programme, which is data about auditability, not a verdict on merit; the mainstream stack's audit lives in the literature rather than a single registry, and its record dwarfs everything below it; and the Rope Programme's precision cell is deliberately annotated to prevent league confusion.

5. Where the Rope Programme genuinely stands (external reviewer's framing, adopted)

CLASSICAL CONTINUUM PHYSICS: the strongest part — a coherent chain from microscopic mechanics through homogenization, electromagnetism, optics, chemistry, and nuclear structure, with explicit boundaries and executable benchmarks. GRAVITY: mixed — the Newtonian mechanism survives with a derivation; classical Einstein gravity is documented as a negative result under current commitments rather than overclaimed, which is scientifically healthier than the original claim. QUANTUM: an explicit boundary — rather than claiming quantum mechanics, the programme identifies where it stops (four of five necessary conditions for a nonlocal completion; the Born rule underived) and what further progress would require. PARTICLE PHYSICS: exploratory — ideas appropriately labeled conjectural or open, with heavier generations out of scope by declaration.

6. The thesis: audit breadth is the metric the field neglects

Evaluations of fundamental theories habitually compare breadth of CLAIM, which is nearly free, rather than breadth of AUDITED claim, which is expensive: it requires deriving numbers, fixing anti-tuning rules in advance, confronting published data, and recording outcomes whichever way they fall. The mainstream stack dominates precisely because its audited breadth approaches its claimed breadth. Among substrate programmes the audit column is almost universally empty. The proposal: adopt the audit column as the primary metric, because it is the only column rhetoric cannot inflate. A programme earns its place in this table by producing numbers that can lose; it earns credibility by keeping the losses on its books. On the day another substrate programme publishes a comparable machine-verified registry with its falsifications intact, this document should be revised — gladly — to reflect it.

Appendix: how to disagree with this document

This document is intended as a framework anyone can use, which means disagreement must be actionable. To dispute any entry, specify: (1) the theory; (2) the column; (3) the evidence — a citation, a computation, or a public artifact; (4) the revised entry you propose; (5) the reason the current entry fails on that evidence. Disputes meeting this form warrant revision of the document; disputes not meeting it are requests for trust, which this document was written to avoid — in both directions. The same protocol applies to the Rope Programme's own cells: every number in Section 3 is recomputable from the shipped registry, and a reader who finds a count wrong should say which claim, which status, and which script disagrees.